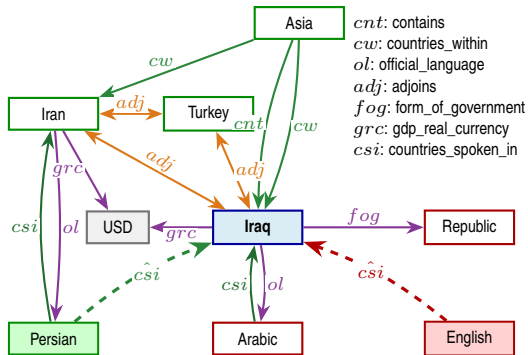


(a) Evidence Motifs Around the Query

query = $\langle ?, \text{countries_spoken_in}, \text{Iraq} \rangle$



(b) Ground Truth: Persian

Rank: 2 $\rightarrow$ 1
Score: 0.288 $\rightarrow$ 0.301

Rules (#Rule=109)

- ϕ_1 : $\text{csi}(X,Y) \leftarrow \text{csi}(X,A) \wedge \text{adj}(A,Y)$ [0.249]
- ϕ_2 : $\text{csi}(X,Y) \leftarrow \text{csi}(X,A) \wedge \text{cw}(B,A) \wedge \text{cnt}(B,Y)$ [0.060]
- ϕ_3 : $\text{csi}(X,Y) \leftarrow \text{ol}(A,X) \wedge \text{adj}(A,B) \wedge \text{adj}(Y,B)$ [0.179]
- ϕ_4 : $\text{csi}(X,Y) \leftarrow \text{ol}(A,X) \wedge \text{grc}(A,B) \wedge \text{grc}(Y,B)$ [0.068]
- ϕ_5 : $\text{csi}(\text{Persian},Y) \leftarrow \text{cnt}(\text{Asia},Y)$ [0.093]

Dependency (#Complementarity=156)

- $d_{1,2}$: $\phi_1 + \phi_2$ [+0.129]
- $d_{3,4}$: $\phi_3 + \phi_4$ [+0.070]
- $\Delta_{\text{dep}} = +0.408$
- adjacency paths and regional paths reinforce Persian

(c) False Positive: English

Rank: 1 $\rightarrow$ 3
Score: 0.300 $\rightarrow$ 0.178

Rules (#Rule=171)

- ϕ_6 : $\text{csi}(\text{English},Y) \leftarrow \text{fog}(Y,\text{Republic})$ [0.194]
- ϕ_7 : $\text{csi}(\text{English},Y) \leftarrow \text{fog}(Y,A)$ [0.253]
- ϕ_8 : $\text{csi}(\text{English},Y) \leftarrow \text{cnt}(\text{Asia},Y)$ [0.222]
- ϕ_9 : $\text{csi}(\text{English},Y) \leftarrow \text{cw}(\text{Asia},Y)$ [0.267]
- ϕ_{10} : $\text{csi}(\text{English},Y) \leftarrow \text{csi}(\text{Arabic},Y)$ [0.318]

Dependency (#Redundancy=201)

- $d_{6,7}$: $\phi_6 + \phi_7$ [-0.038]
- $d_{8,9}$: $\phi_8 + \phi_9$ [-0.018]
- $\Delta_{\text{dep}} = -0.122$
- government and continent cues overlap for English